

Business Proposal Letter

January 7, 2026

TO: THE SCIENCE FACULTY

Southcom National High School (SNHS)

SUBJECT: Proposal for Digital Laboratory Transformation & Pilot Testing Invitation

Dear Fellow Science Educators,

As an alumnus of **Southcom National High School**, I am well aware of the daily challenges we face in our laboratories: the missing beakers that never get replaced, the struggle of identifying "freeloaders" in group work, and the hours wasted on manual logbooks.

I am proposing a partnership to implement a **Hybrid Laboratory Inventory & Collaborative LMS**, a system I have developed specifically to solve these manual lab headaches through automation, real-time tracking, and data-driven accountability.

1. The Problem-Solution Statement

The current reliance on manual processes in the laboratory setting leads to significant financial loss and administrative burden. The proposed system directly addresses these issues through digital innovation.

The Manual Struggle	The Digital Solution
Asset Leakage: ₱30k–₱50k of equipment is lost or broken annually due to paper logbooks that fail to enforce individual accountability.	Distributed Liability: QR-based borrowing that links every item to the specific student holding it, not just the group leader.
Grading Guesswork: Teachers struggle to prove who actually worked in a group and who simply "signed the paper."	Forensic Heatmaps: A dashboard showing the exact percentage of effort, word count, and time contributed by each student.

The Manual Struggle

Administrative Burnout: Teachers spend 8–17 hours/week on non-teaching tasks (inventory updates, paper reports).

The Digital Solution

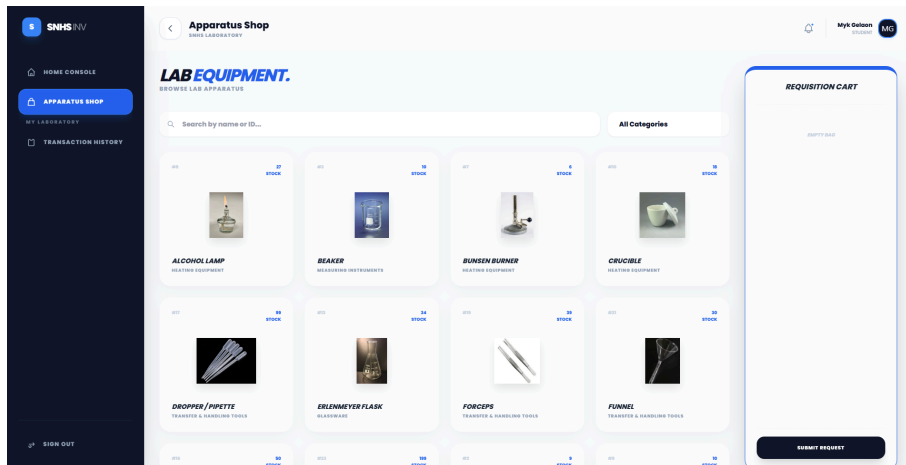
Process Automation: Instant stock updates and digital submission loops that reduce administrative work by an estimated 60%.

2. System Workflow: The Smart-Lab Engine

The system is designed as a complete **Laboratory Ecosystem** that handles both the physical logistics of equipment and the academic output of students.

A. Logistics and Inventory Engine

- **The Borrowing Wizard:** Students select apparatus via a "Shopping Cart" interface. The system performs an instant inventory check against `Available_Qty` and generates a digital Borrowing Slip with a QR Code.



SNHS LAB

BORROWING RECORD
2926-01-07

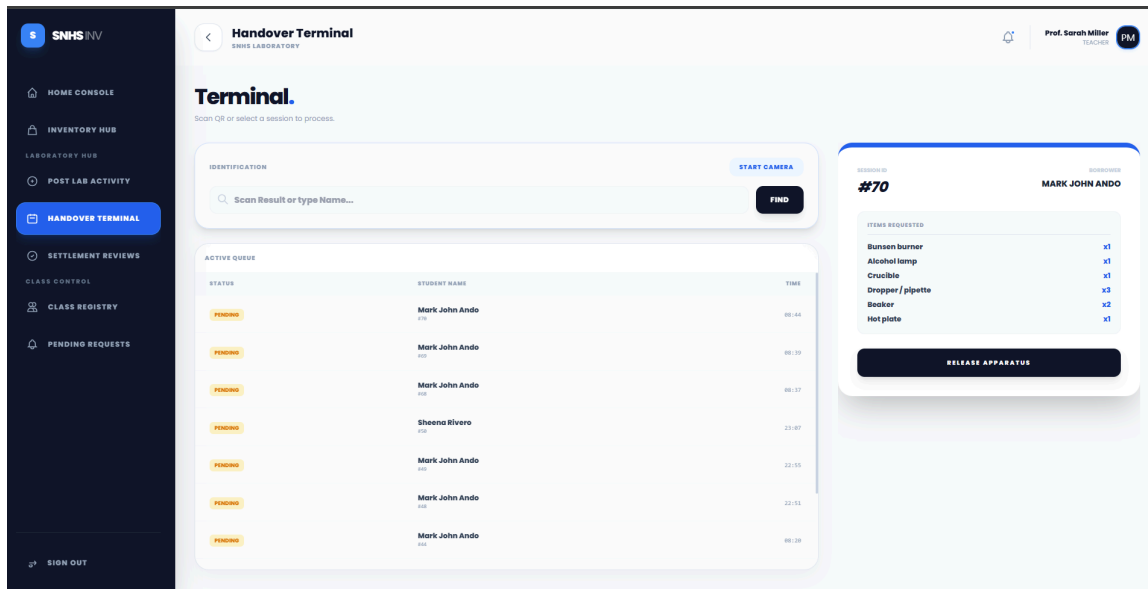
STUDENT: MARK JOHN ANDO
ID: #78
ACTIVITY: TRANSCENDENT RESEARCH

ITEM	QTY
Bunsen burner	1
Alcohol lamp	1
Crucible	1
Dropper / pipette	3
Beaker	2
Hot plate	1

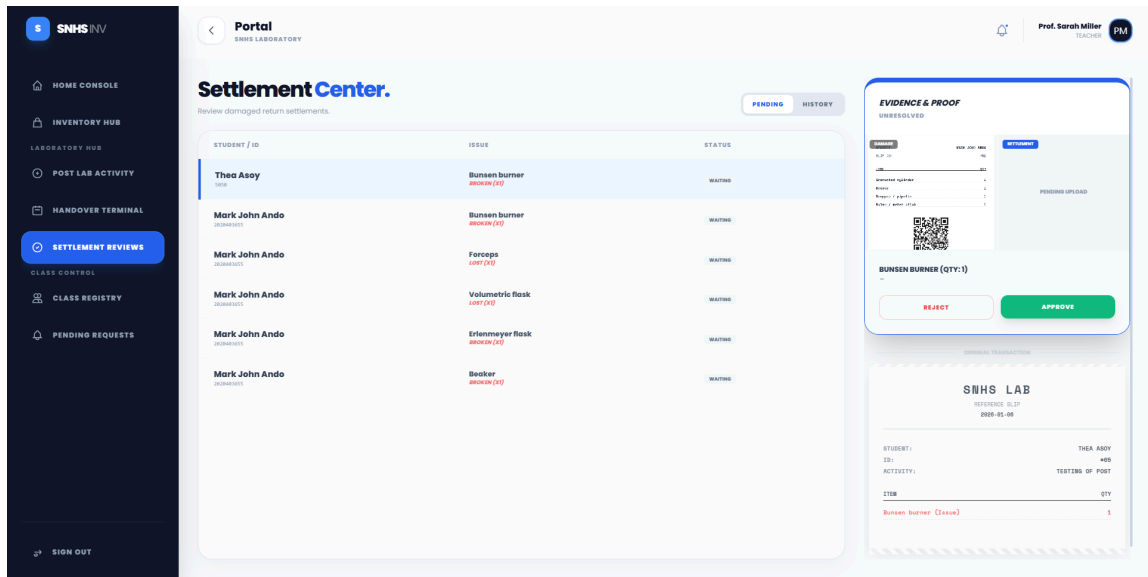


PENDING

- **Approval Workflow:** Teachers act as gatekeepers, reviewing student requests via a digital dashboard to approve or deny them before items are released.

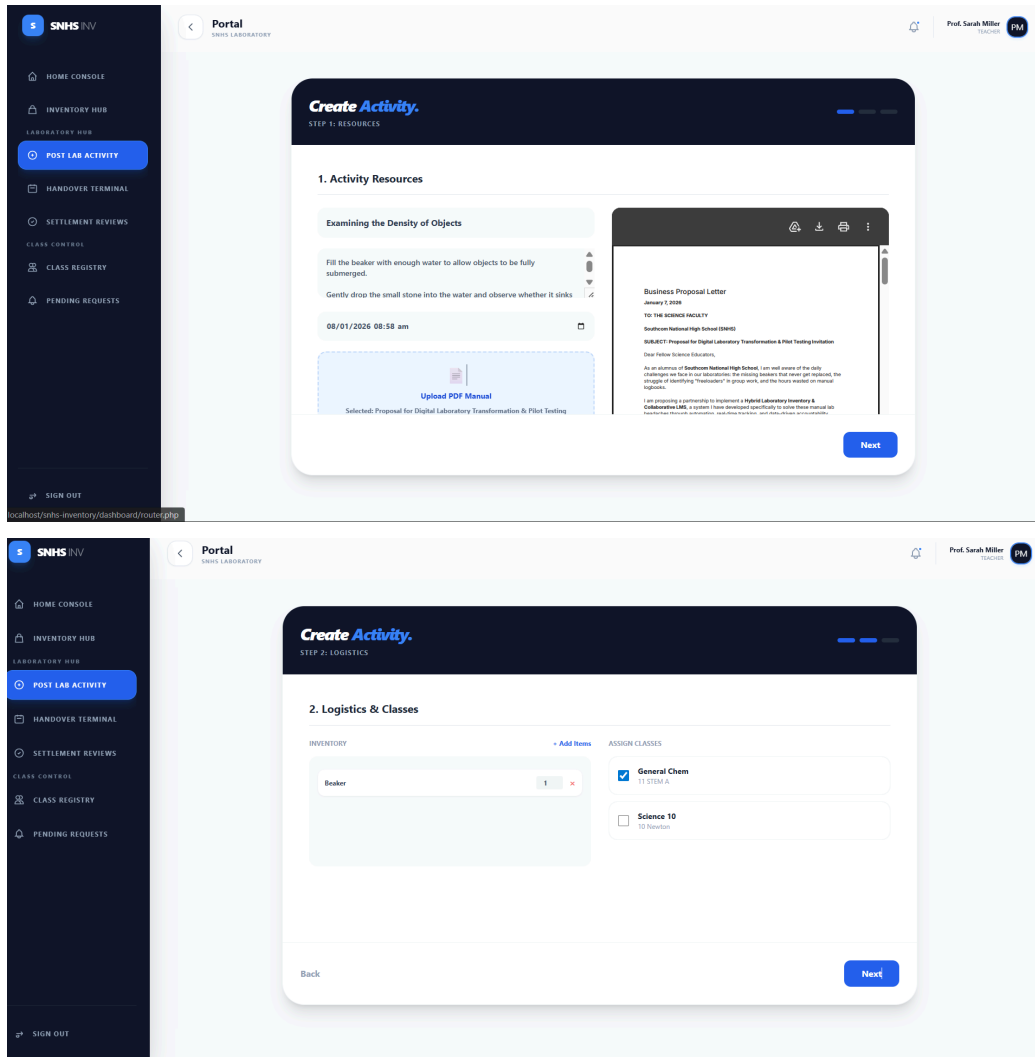


- **Liability Management:** Students upload visual proof (photos) of returned or broken items. Liabilities are tied to individual **UserIDs** within a **SessionID**, ensuring the correct student is held accountable for school assets.



B. Activity Posting Done by Teacher

- **Smart Resource Setup:** Teachers create activities by uploading digital manuals and selecting specific lab apparatus from a "live" inventory. This ensures that students only request what is physically available in the stockroom before the experiment begins.



- **Flexible Grouping Strategies:** The system offers three distinct modes to fit your teaching style:
 - **Individual Mode:** Every student works and is graded independently.

Create Activity.
STEP 3: SETTINGS

3. Workflow Logic

Mode

INDIVIDUAL

GROUP

Submission

☒ File Upload

☐ Smart-Lock Builder

Back

Publish Activity

- **Student Choice:** Students form their own teams through a "Group Lobby," perfect for fostering leadership and social learning.

Team Lobby.
MANAGE YOUR SQUAD

TEAM NAME

Group1

SELECT MEMBERS 0/2

☐ Jomar Jun

☐ Thea Asoy

☐ Myk Gelson

☐ Awis Barcelona

* YOU ARE THE LEADER.

CREATE TEAM

- **Smart Auto-Assign:** The system automatically shuffles the class into balanced teams based on your preferred group size.

The screenshot shows the 'Create Activity' interface, specifically 'STEP 3: SETTINGS'. The main heading is '3. Workflow Logic'. There are two main sections: 'Mode' and 'Submission'.

Mode: Includes a 'Mode' section with 'INDIVIDUAL' and 'GROUP' buttons. The 'GROUP' button is selected. Below this is a 'STRATEGY' dropdown menu set to 'Smart Auto-Assign'. At the bottom is a 'GROUP LIMIT' input field with the value '3'.

Submission: Includes a 'Submission' section with two radio buttons: 'File Upload' (selected) and 'Smart-Lock Builder'.

At the bottom left is a 'Back' button, and at the bottom right is a 'Publish Activity' button.

- **Manual Grouping Board:** For high-stakes experiments, you can use a "Drag-and-Drop" board to personally assign students to specific teams and designate Group Leaders. This ensures that every group is balanced and has a responsible student-in-charge before the activity is published.

The screenshot shows the 'Grouping Board' interface. At the top, it says 'Grouping Board.' and 'TARGET SIZE: 3 | CLASS SIZE: 7'. There are 'CANCEL' and 'CONFIRM & PUBLISH' buttons at the top right.

CLASS ROSTER: A list of students on the left: 'Myk Gelson', 'Shenna Rivers', and 'Thera Avey'.

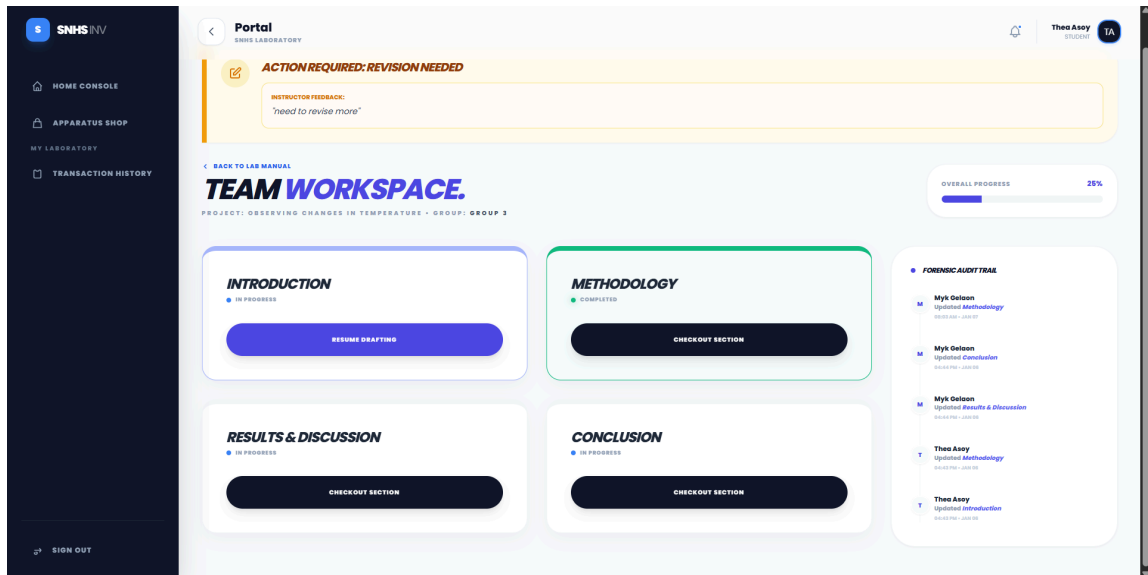
Groups: Three groups are shown on the right:

- Group 1:** Contains 'Amita Barakoma' and 'Jamar Ben'. A button 'Click to add Myk Gelson' is at the bottom.
- Group 2:** Contains 'Samuel Mokuail'. A button 'Click to add Myk Gelson' is at the bottom.
- Group 3:** Contains 'Mark John Ando'. A button 'Click to add Myk Gelson' is at the bottom.

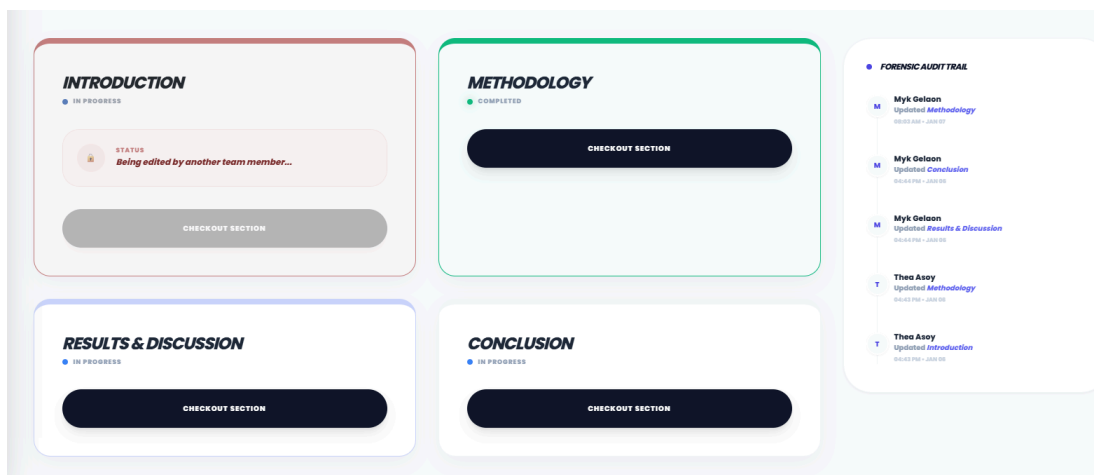
A dashed box in the center says '+ ADD EXTRA GROUP'.

C. Collaborative Workspace and Forensic Engine

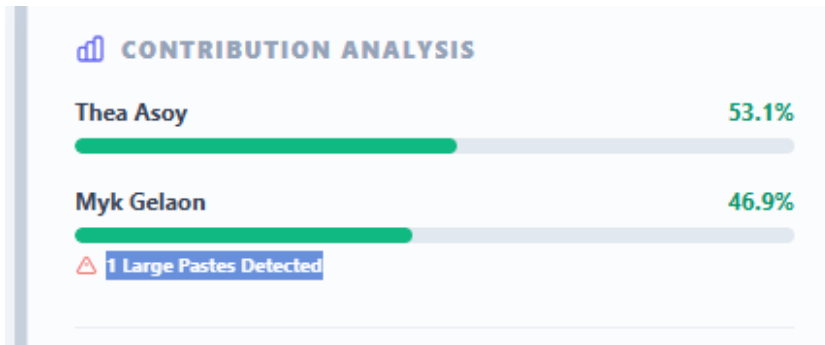
- **Section-Based Editing:** Reports are broken into modular cards (Introduction, Methodology, Results, etc.).



- **Concurrency Locking:** I have implemented a custom **Lock Manager**. When one student opens a section, it locks for all others in real-time to prevent overwriting conflicts.



- **The Forensic Engine:** The system calculates "Net Contribution" (characters added vs. deleted). It includes **Paste Detection** to flag text injections (e.g., >500 characters in one second), alerting teachers to potential AI or Wikipedia copy-pasting.



- **Revision Workflow:** Teachers can "Return" work for revision. This triggers a state change that unlocks the workspace and displays a "Revision Needed" banner with instructor feedback.

← BACK TO DASHBOARD **Grading Console** TARGET: GROUP 3 ESTIMATED LENGTH: 134 Words **RETURNED**

FINAL REPORT PREVIEW

Observing Changes in Temperature

Group 3

Introduction

Temperature changes occur when substances with different temperatures are combined. This activity demonstrates how heat is transferred when warm and cold water are mixed.

Methodology

Warm water was placed in a beaker and its initial temperature was measured using a thermometer. Cold water was then added and the mixture was stirred. The final temperature was recorded.

Results & Discussion

After mixing the warm and cold water, the temperature of the mixture was observed to be between the initial temperatures of the two liquids.

Conclusion

CONTRIBUTION ANALYSIS

Student	Score
Thea Asoy	53.1%
Myk Gelaon	46.9%

☒ Auto-adjust grades by effort?

BASE GROUP GRADE

00 / 100

TEACHER FEEDBACK

Write helpful feedback for the group here...

ADJUST INDIVIDUAL SCORES

Student	Score
Thea Asoy	53.1% (50.00-100.00)
Myk Gelaon	46.9% (50.00-100.00)

POST FINAL GRADES

RETURN FOR REVISION

3. Flexible Deployment Options

The system offers two deployment options based on the department's needs and resource availability:

Option	How it Works	Requirements/Notes
Option 1: Local-Host Setup (No Internet Required)	A faculty member's personal laptop can be used as the central host. Students connect to the laptop's signal via Wi-Fi.	Works offline , ensuring 100% uptime regardless of internet stability.

Option	How it Works	Requirements/Notes
Option 2: Online Cloud Hosting	Hosted on a high-speed Singapore-based server for access anywhere (school or home).	Requires an annual hosting/domain fee (approx. ₱7,500) to be covered by the department.

4. Pricing & Professional Commitment

I am currently a **Second-Year student graduating from the Associate in Computer Technology (ACT) program, majoring in Application Development**. Because this is my **first-ever professional business deal**, I am offering my services at a rate significantly below standard market value to build my portfolio while giving back to my Alma Mater.

The investment required is detailed in the table below:

Service Component	Corporate Agency Rate	SNHS Alumnus Rate
Custom PHP/MySQL Engine	₱150,000+	₱15,000
QR & Lock-Manager API	₱40,000+	INCLUDED
Data Digitization (Labor)	₱15,000	INCLUDED
TOTAL INVESTMENT	₱205,000+	₱15,000

5. CALL TO ACTION: THE 1-DAY BETA TEST

I am requesting a Safety Check to prove the system's value in a live environment before any commitment is made. This session will serve as a full tutorial and guided walkthrough of the platform's capabilities.

Pilot Testing Details:

Detail	Information
Date:	January 12, 2026 (Monday)
Target Class:	11 STEM
Activity:	Full System Tutorial and Guided Walkthrough

I will be on-site to handle the local server setup using a personal laptop to host the session. During this walkthrough, I will guide both faculty and students through the end-to-end process, from generating QR slips to analyzing forensic contribution heatmaps. Your only role is to observe how the system manages your students and provides you with objective grading data in real-time.

Are you ready to witness the transformation of the SNHS laboratory this coming Monday?

Sincerely,

mark ando

ACT - Application Development | SNHS Alumnus